TATA POWER LTD.

Equity Research & Valuation — Financial Analysis Report

Integrated Financial Model: Statement Analysis, DCF, Relative Valuation, Utility Credit Scorecard & Risk Analysis

Methodology: Informed by Aswath Damodaran's (NYU Stern) Valuation Frameworks, Adapted to Available Data

Educational Portfolio Project — Not Investment Advice

Contents

1. Executive Summary
2. Company Overview
3. Historical Financial Performance
3A. Financial Statement Analysis
4. Ratio Analysis & DuPont Decomposition
5. Risk Profile — Beta & Cost of Equity
6. Weighted Average Cost of Capital
7. Utility Credit Quality Scorecard
8. Altman Z-Score — Legacy Cross-Reference
9. Discounted Cash Flow Valuation
10. Relative Valuation
11. Value-at-Risk Analysis
12. Football Field Synthesis
13. Limitations & Disclosures
14. Conclusion

1. Executive Summary

This report documents a comprehensive equity research and valuation project on Tata Power Company Ltd., India's largest integrated power utility. The underlying Excel workbook (25 sheets) draws on several of Aswath Damodaran's valuation frameworks — applied where the available data supported them, and adapted or approximated where it did not — across ten years of historical financial statement analysis (FY2017–FY2026), ratio and DuPont decomposition, CAPM-based cost of equity, WACC construction, a three-stage discounted cash flow (DCF) model, peer-based relative valuation, a custom 9-driver Utility Credit Quality Scorecard, Value-at-Risk analysis across three methodologies, and a football field valuation synthesis. This is an approximation of Damodaran's approach built with publicly available data, not a claim of full methodological replication.

This project was built for educational and portfolio purposes and does not constitute investment advice. A defining feature of this analysis is that it does not smooth over its own limitations: where a model produces a result at odds with observable reality **— the Altman Z-Score's distress signal against Tata Power's investment-grade rating**, or **a DCF value far below market price —** that divergence is investigated, explained, and disclosed rather than adjusted away.

Key Figures at a Glance

Metric	Value
Return on Equity (ROE)	15.05%
Beta (2-yr weekly regression vs. Nifty 50)	1.26
Cost of Equity (CAPM)	15.52%
WACC	11.78%
DCF Equity Value per Share	₹19.17
Market Price per Share	₹402.35
Utility Credit Scorecard (FY26)	38 / 100 — "High Risk"
Actual CRISIL Rating (FY26)	AA+ / Stable

2. Company Overview

Tata Power is India's largest integrated power company, with operations spanning generation, transmission, and distribution across India, Georgia, and Zambia. The company operates over 14,700 MW of diversified generation capacity across thermal, hydro, solar, and wind assets, and holds a publicly stated target of reaching 70% clean energy capacity by 2030 — a transition that is the central driver of its capital expenditure cycle and, by extension, several of the valuation results in this report.

Business Segments

- Regulated Generation — thermal and hydro assets operating under India's cost-plus regulated tariff framework
- Transmission & Distribution — including Mumbai and Delhi (Tata Power Delhi Distribution) distribution licenses
- Renewable Energy — Tata Power Renewable Energy Ltd. (TPREL), not separately listed
- International Assets — generation and mining interests in Georgia and Zambia

Credit Standing

Tata Power holds long-term credit ratings of CRISIL AA+/Stable and ICRA AA+/Stable. The CRISIL rating has moved from AA- (FY2017) to AA (Nov 2020) to AA+ (Apr 2024), a clean upgrade trajectory with zero downgrades over the full ten-year model period. Short-term ratings (commercial paper) have held at CRISIL A1+, the highest short-term category, throughout.

Shareholding Pattern (FY26)

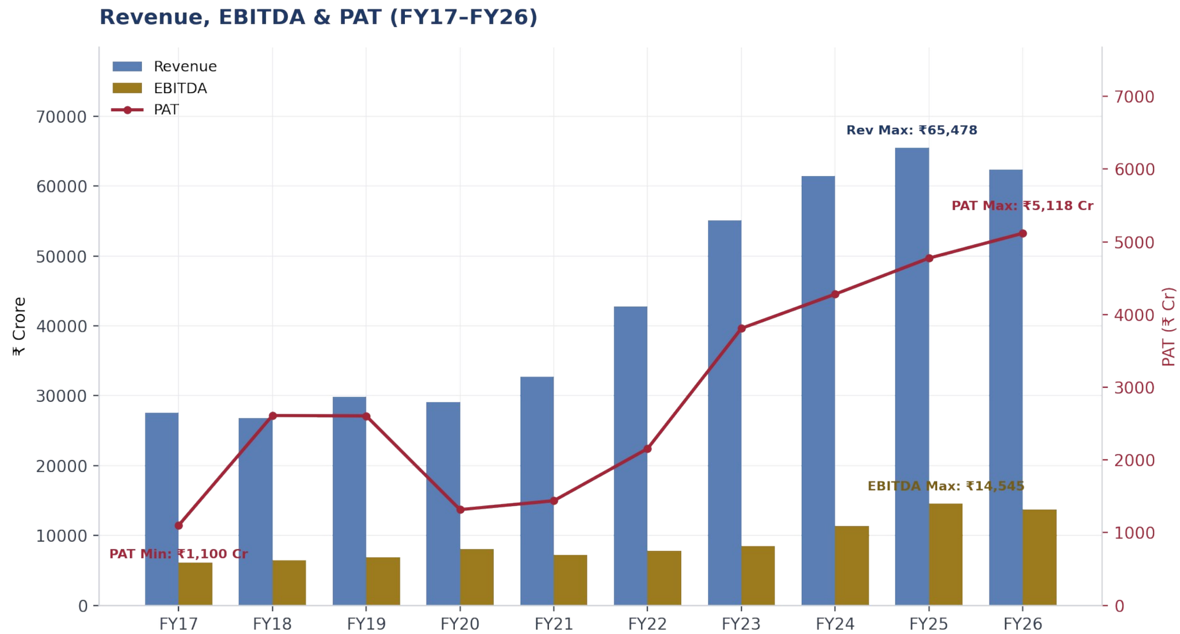
Category	% Holding
Promoters (Tata Sons and affiliates)	46.86%
Domestic Institutional Investors (DIIs)	17.98%
Foreign Institutional Investors (FIIs)	10.04%
Government	0.31%
Other Public Shareholders	24.81%

Top individual holders include Tata Sons Private Limited (45.21%), LIC (5.18%), Nippon Life India Asset Management (2.25%), and BlackRock, Inc. (1.79%).

The concentrated promoter holding (46.86%, almost entirely Tata Sons) is a relevant qualitative factor for the credit analysis in Section 7: parent-group support from the Tata conglomerate is one of the qualitative factors real rating agencies weight that this project's purely quantitative scorecard does not capture.

3. Historical Financial Performance

Revenue grew from ₹27,588 Cr in FY17 to a peak of ₹65,478 Cr in FY25, before declining to ₹62,429 Cr in FY26. EBITDA grew even faster in percentage terms — from ₹6,093 Cr to a peak of ₹14,545 Cr in FY25 — reflecting margin recovery as the generation and distribution businesses stabilized and renewable capacity additions began contributing. Net Profit (PAT) shows a more volatile path, more than doubling from FY17 to FY18 before a multi-year rebuild through FY26.

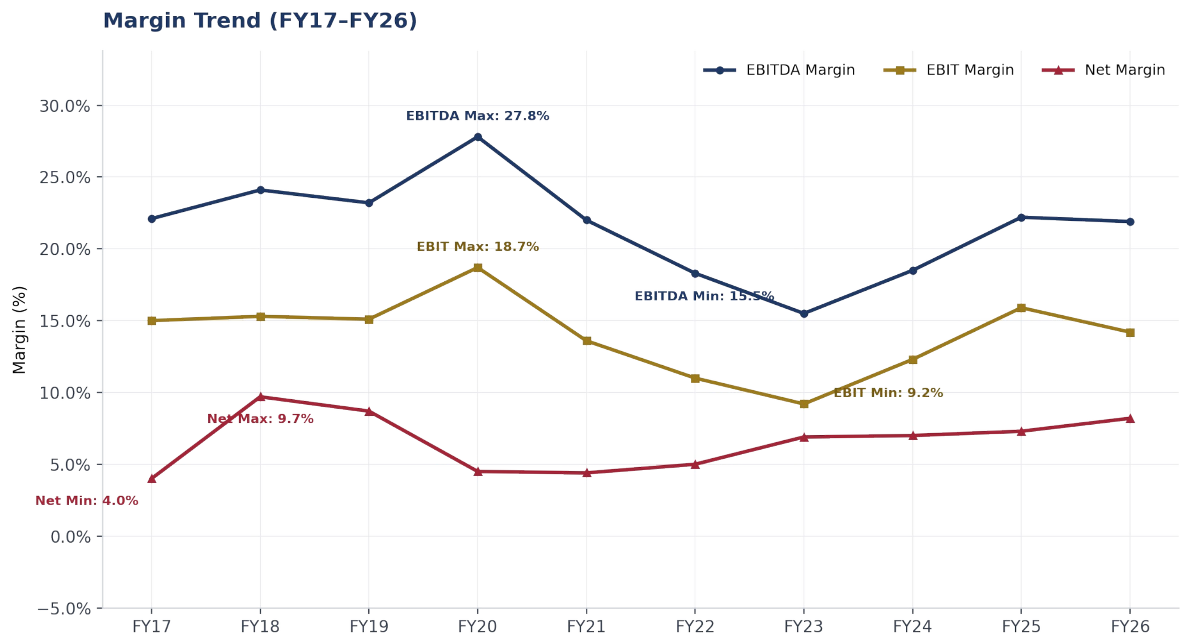


Fiscal Year	Revenue (₹ Cr)	EBITDA (₹ Cr)	EBIT (₹ Cr)	PAT (₹ Cr)	EPS (₹)
FY17	27,588	6,093	4,137	1,100	4.07
FY18	26,840	6,464	4,118	2,611	9.65
FY19	29,881	6,918	4,525	2,606	9.63
FY20	29,136	8,096	5,462	1,316	4.87
FY21	32,703	7,194	4,449	1,439	4.50
FY22	42,816	7,846	4,724	2,156	6.75
FY23	55,109	8,532	5,093	3,810	11.92
FY24	61,449	11,360	7,574	4,280	13.39
FY25	65,478	14,545	10,428	4,775	14.94
FY26	62,429	13,692	8,881	5,118	16.02

Source: Historical Financial Statements sheet, derived from Screener.in company filings.

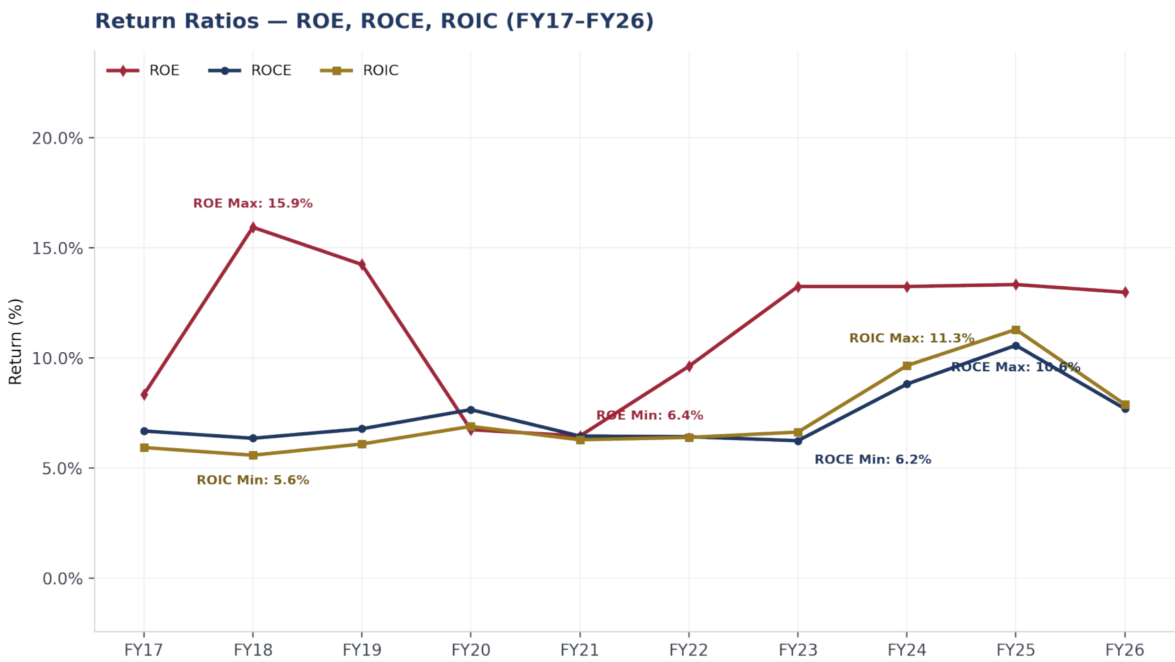
The FY22–FY23 period shows marked EBITDA and EBIT margin compression despite strong revenue growth — consistent with a period of rising fuel/input costs across the Indian power sector that were not immediately passed through to regulated tariffs. Margins recovered from FY24 onward as cost pass-through normalized and higher-margin renewable capacity scaled. Notably, EPS has grown far more consistently than PAT alone would suggest across FY21–FY26, aided by the fixed share count over that window.

Margin Trend



Return Ratios — ROE, ROCE, ROIC

Fiscal Year	ROE	ROCE	ROIC
FY17	8.3%	6.7%	5.9%
FY18	15.9%	6.3%	5.6%
FY19	14.2%	6.8%	6.1%
FY20	6.7%	7.6%	6.9%
FY21	6.4%	6.4%	6.3%
FY22	9.6%	6.4%	6.4%
FY23	13.2%	6.2%	6.6%
FY24	13.2%	8.8%	9.6%
FY25	13.3%	10.6%	11.3%
FY26	13.0%	7.7%	7.9%

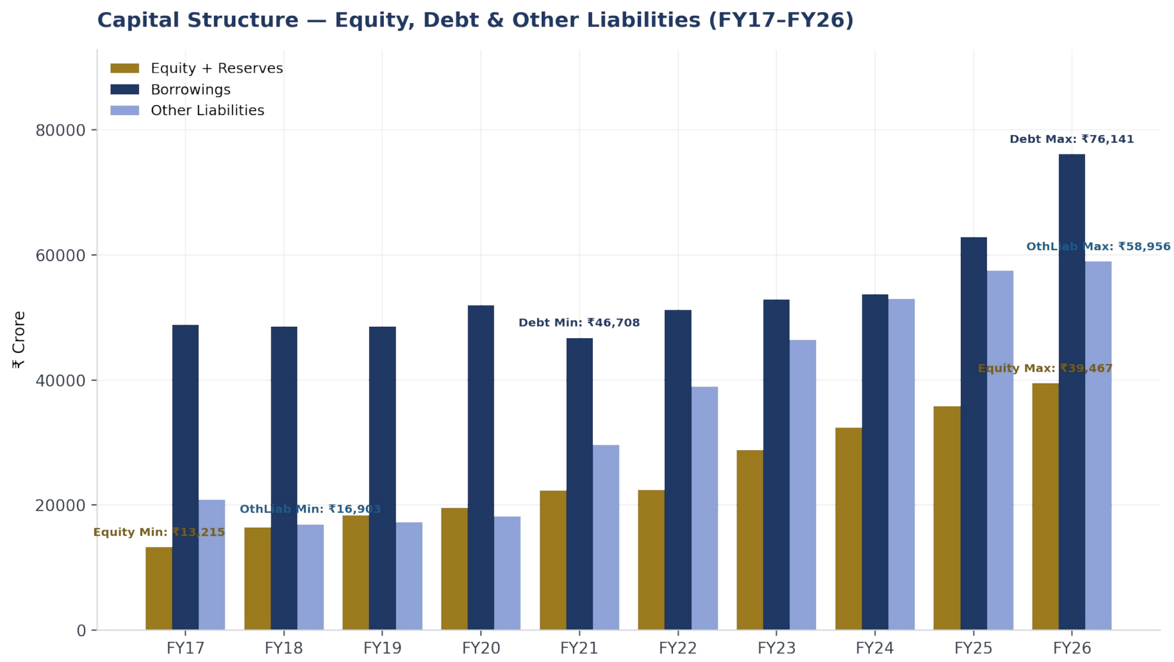


ROIC has run consistently below ROE across the full period — expected for a leveraged business — but the gap narrowed markedly in FY24–FY25 as operating returns improved, before both compressed again in FY26 alongside the debt increase discussed in Section 7.

3A. Financial Statement Analysis

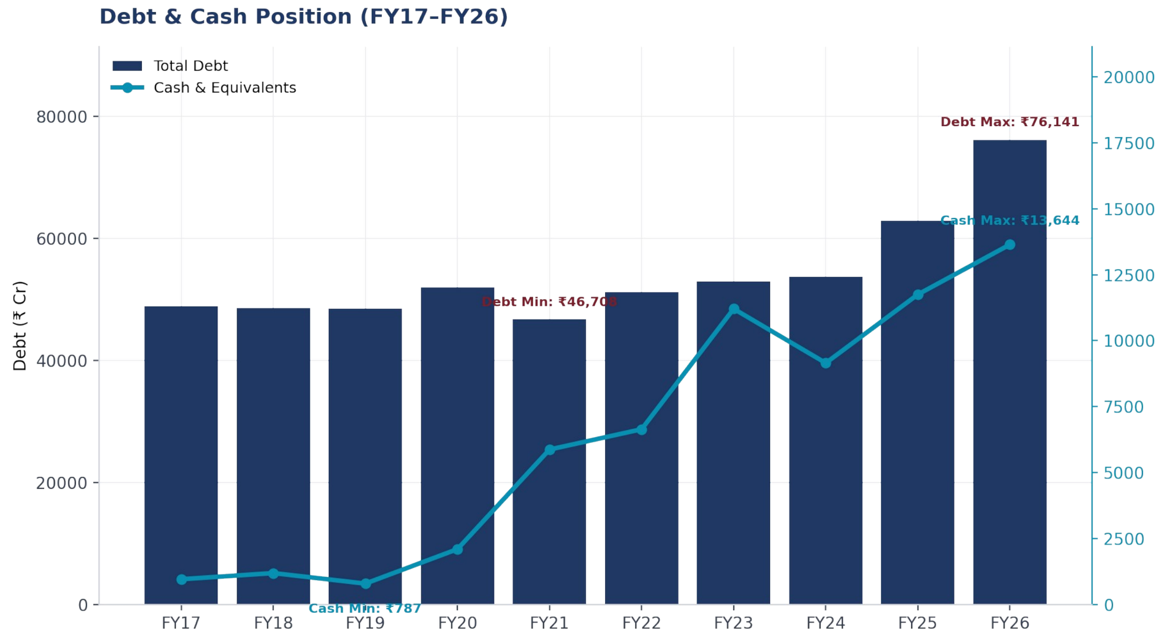
Capital Structure

Tata Power's balance sheet has grown from ₹82,829 Cr to ₹174,564 Cr in total assets/liabilities over the model period, funded by a mix of retained earnings, borrowings, and other liabilities (including regulatory and deferred tax balances).



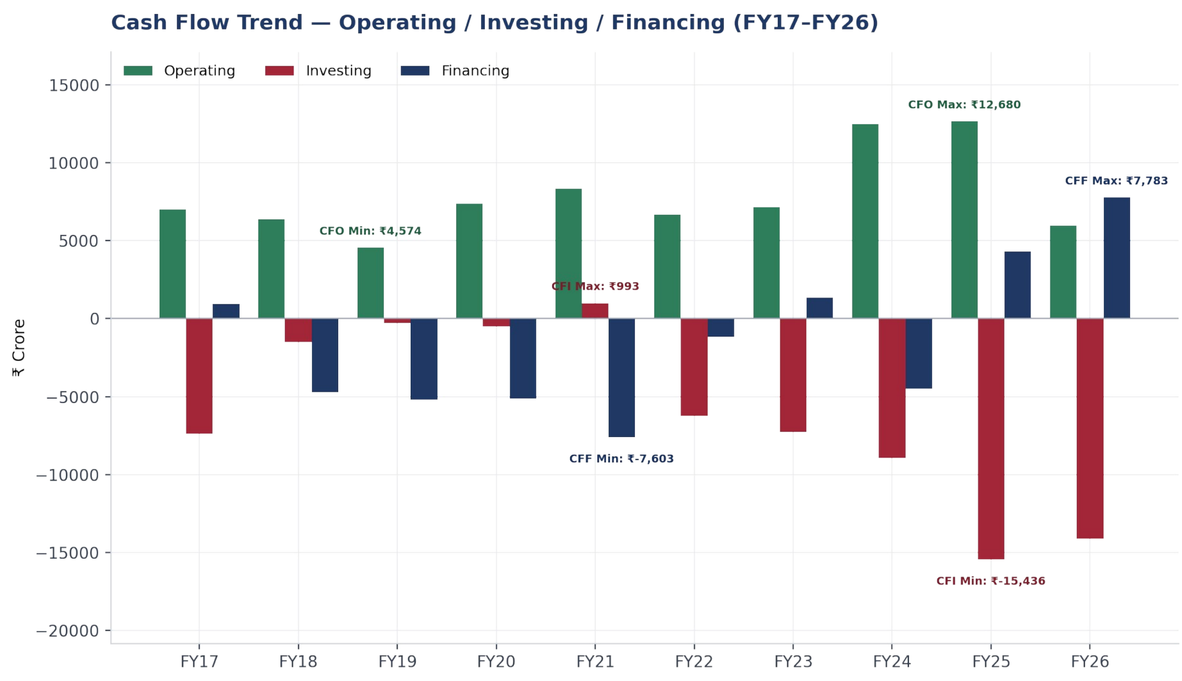
Debt & Cash Position

Total borrowings rose from ₹48,815 Cr (FY17) to ₹76,141 Cr (FY26), with the steepest single-year increase in FY26 (+21% YoY) coinciding with the accelerated renewable capex cycle discussed in Section 7. Cash and equivalents have grown broadly in step, providing partial offset.



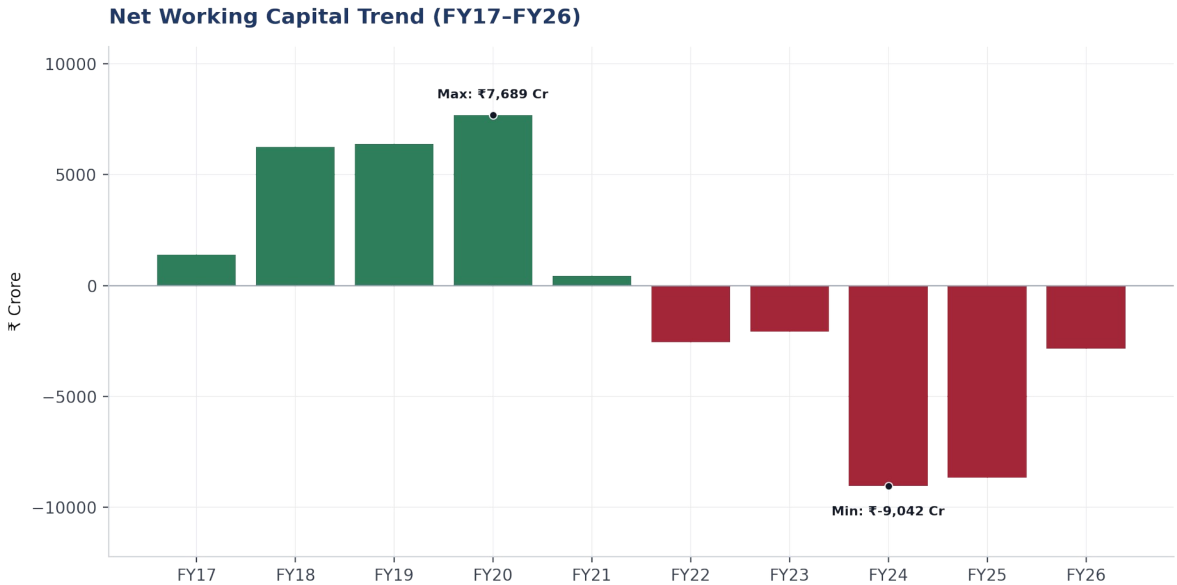
Cash Flow Trend

Operating cash flow has remained consistently positive throughout, though it dipped sharply in FY26 (₹5,993 Cr, down from ₹12,680 Cr in FY25) alongside the working capital swing noted below. Investing cash outflows have grown substantially since FY22, reflecting the renewables capex program, funded increasingly by financing inflows (positive financing cash flow in FY25 and FY26, a reversal from the net debt-paydown pattern of FY18–FY21).



Working Capital Trend

Net working capital swung from consistently positive through FY20 (peaking at ₹7,689 Cr) to persistently negative from FY21 onward, reaching -₹9,042 Cr in FY24 before partially recovering to -₹2,854 Cr in FY26. A negative working capital position is common and often structurally healthy for utilities with regulatory payables/recoverables, but the magnitude and volatility of this swing is a direct contributor to the volatile reinvestment-rate series used in the DCF (Section 9).


Net Working Capital Trend (FY17–FY26)
₹ Crore
10000
5000
0
−5000
−10000
Max: ₹7,689 Cr
Min: ₹-9,042 Cr
FY17
FY18
FY19
FY20
FY21
FY22
FY23
FY24
FY25
FY26

4. Ratio Analysis & DuPont Decomposition

The DuPont framework decomposes Return on Equity into three multiplicative components — net profit margin, asset turnover, and financial leverage (equity multiplier) — isolating which lever is driving shareholder returns.

Component	Value	Formula
Net Profit Margin (A)	9.73%	Net Profit / Revenue
Asset Turnover (B)	0.32x	Revenue / Average Total Assets
Equity Multiplier (C)	4.78x	Average Total Assets / Average Equity
Return on Equity (A × B × C)	15.05%	Net Profit / Average Equity

The decomposition shows that Tata Power's ROE is driven substantially more by **financial leverage (a 4.78x equity multiplier)** than by operating profitability (a 9.73% net margin, 0.32x asset turnover). This is structurally typical of capital-intensive regulated utilities, which fund a large, long-lived asset base predominantly with debt — but it also means **ROE is more sensitive to interest rate and leverage changes than to operating efficiency gains**, a consideration directly relevant to the credit risk analysis in Section 7.

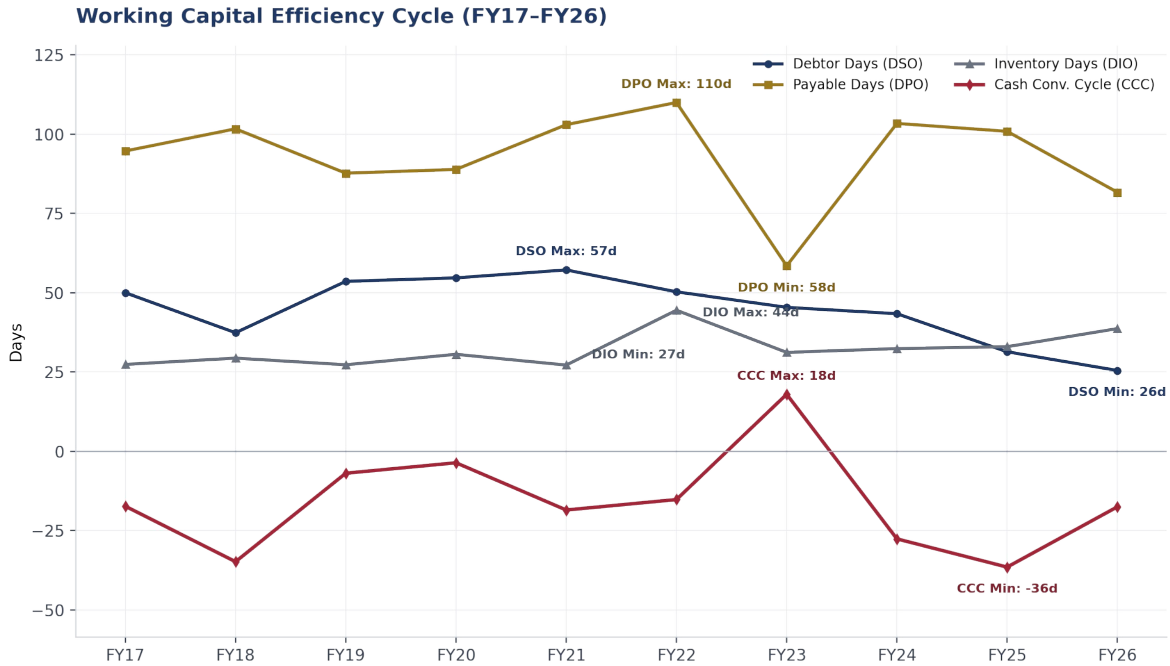
Cross-check: Return on Assets

Return on Assets (Net Profit Margin × Asset Turnover) computes to 3.15%, confirming the DuPont identity holds algebraically across the full decomposition (3.15% × 4.78x equity multiplier = 15.05% ROE).

Efficiency Ratios — Working Capital Cycle

Beyond the DuPont components, the working capital efficiency cycle provides a complementary view of operational management, tracking how quickly the company converts receivables and inventory into cash relative to how it manages payables.

Fiscal Year	Debtor Days	Payable Days	Inventory Days	Cash Conv. Cycle
FY17	50.0	94.7	27.4	-17.3
FY18	37.4	101.7	29.4	-34.8
FY19	53.6	87.7	27.3	-6.9
FY20	54.7	88.9	30.6	-3.6
FY21	57.2	103.0	27.2	-18.5
FY22	50.3	110.0	44.5	-15.2
FY23	45.4	58.5	31.2	18.0
FY24	43.4	103.4	32.4	-27.6
FY25	31.4	100.9	33.0	-36.5
FY26	25.5	81.7	38.7	-17.5



Tata Power's Cash Conversion Cycle has been **negative in 9 of 10 years** — the company is typically paid by customers before it needs to pay its own suppliers, a structurally favorable working-capital position common among utilities with regulated, relatively fast-cycle receivables. **Debtor Days have improved markedly and consistently since FY21 (57.2 days → 25.5 days)**, suggesting genuine collections efficiency gains rather than a one-off improvement — a positive operating trend not otherwise visible in the margin or leverage metrics.

5. Risk Profile — Beta & Cost of Equity

Beta was estimated via ordinary least squares regression of Tata Power's weekly stock returns against the Nifty 50 index over a 2-year (104-week) window — a standard convention balancing statistical power against staleness in a fast-evolving capital structure.

Regression Output

Statistic	Value
Beta (slope coefficient)	1.26493
Standard Error (Beta)	0.16649
t-Statistic (Beta)	7.60
P-value (Beta)	1.50×10^{-11}
R^2	0.3614
Adjusted R^2	0.3552
Correlation Coefficient	0.6012
F-Statistic (ANOVA)	57.73
Degrees of Freedom (Residual)	102
Observations	104 weeks

The Beta estimate is **highly statistically significant ($p < 0.001$, t-statistic of 7.60)**, giving confidence in its use for the cost of equity calculation that follows — a meaningful contrast to the regression-based relative valuation in Section 10, which **is not statistically significant** and is treated accordingly.

Cost of Equity Build-Up (CAPM)

Component	Value
Risk-Free Rate (India 10Y G-Sec proxy)	6.80%
Equity Risk Premium (rating-based)	7.08%
Unlevered Beta	0.894
Levered Beta (at target capital structure)	1.231
Cost of Equity = $R_f + \beta \times ERP$	15.52%

The Equity Risk Premium uses Damodaran's rating-based approach (India's sovereign default spread applied to a mature-market ERP), rather than a trailing historical-returns approach. This choice was deliberate: the historical-returns-implied ERP for India over the sample period computes to a negative figure — a well-documented artifact of short-window historical volatility in emerging markets — which would be economically invalid to use directly in CAPM.

6. Weighted Average Cost of Capital

WACC blends the after-tax cost of debt and the cost of equity, weighted by each source's share of the target capital structure.

Component	Value
Pre-Tax Cost of Debt	6.90%
Tax Rate	30%
Post-Tax Cost of Debt	4.83%
Cost of Equity	15.52%
Target Weight — Debt	35%
Target Weight — Equity	65%
WACC = (Wd × Kd) + (We × Ke)	11.78%

The target capital structure (**35% debt / 65% equity**) is set close to Tata Power's current market-value leverage (debt at 37.2% of total capital, using market capitalization for the equity component), rather than an arbitrarily chosen or theoretical target — a deliberate methodological choice, since Tata Power's actual observed market-value leverage has fluctuated considerably over the model period (from over 2.0x debt/equity in FY17 to a low of 0.43x in FY24, before rising again to 0.63x by FY26) and does not exhibit a single stable long-run trend that would justify a materially different target.

7. Utility Credit Quality Scorecard

Standard distress-prediction models — most notably the Altman Z-Score (Section 8) — are calibrated on manufacturing-sector failure data and are known to systematically misclassify capital-intensive, regulated utilities as financially distressed. Two of the Altman model's five inputs are structurally unfavorable for this sector: asset turnover (naturally low given a large, long-lived fixed-asset base relative to annual revenue) and leverage (structurally high but low-risk, given predictable regulated cash flows). This section develops a custom, utility-specific scorecard as a methodologically appropriate replacement.

Scorecard Design

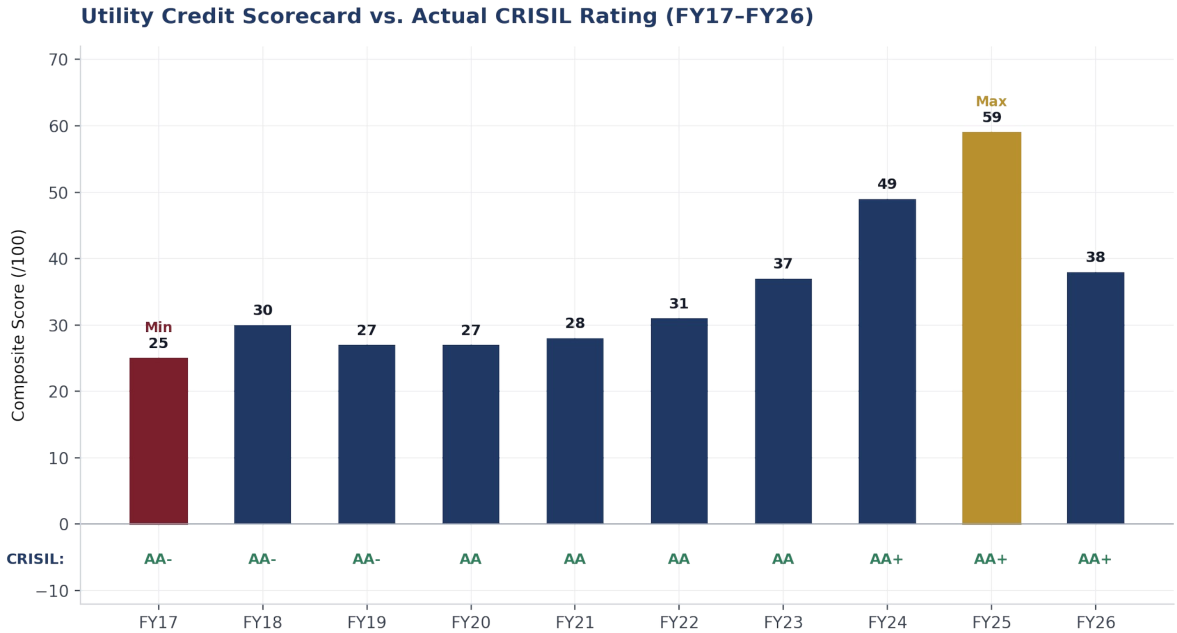
Nine drivers were selected, each scored against bands calibrated to Indian utility-sector norms rather than global developed-market thresholds, and combined into a 100-point composite:

Driver	Max Score	Rationale
Interest Coverage	25	Direct debt-servicing capacity from current earnings
Net Debt / EBITDA	20	Primary leverage metric used by CRISIL/ICRA for Indian utilities
FFO / Net Debt	15	Moody's-style cash-flow leverage metric
ROIC	10	Ties credit quality to the same growth engine driving the DCF
Operating Cash Flow / Debt	10	Independent cash-generation cross-check
Debt / Capitalization	5	Structural leverage, India-recalibrated bands
Dividend Payout Ratio	5	Cash retention / deleveraging capacity
Current Ratio	5	Minor liquidity sanity check
Altman Z-Score	5	Down-weighted legacy cross-reference

Full 10-Year Driver Detail

Driver	FY17	FY18	FY19	FY20	FY21	FY22	FY23	FY24	FY25	FY26
ROIC	5.9%	5.6%	6.1%	6.9%	6.3%	6.4%	6.6%	9.6%	11.3%	7.9%
Interest Coverage	1.23x	1.09x	1.09x	1.22x	1.11x	1.22x	1.16x	1.63x	2.22x	1.69x
Net Debt/EBITDA	8.16x	7.64x	7.21x	6.44x	6.08x	6.14x	5.52x	4.45x	3.98x	5.15x
FFO/Net Debt	6.0%	9.8%	9.9%	7.3%	8.4%	9.6%	12.4%	13.5%	12.8%	11.8%
OCF/Debt	13.8%	12.6%	9.0%	13.6%	16.8%	12.2%	12.3%	21.0%	18.2%	7.1%
Current Ratio	1.17x	1.56x	1.57x	1.63x	1.13x	1.03x	1.08x	0.93x	0.96x	1.10x
Debt/Cap	67.5%	70.3%	71.7%	85.9%	60.1%	41.8%	49.0%	32.1%	36.7%	41.0%
Payout Ratio	32.0%	13.5%	13.5%	31.8%	34.4%	25.9%	16.8%	14.9%	15.1%	15.6%

Composite Score vs. Actual Credit Rating



Fiscal Year	Composite Score	Scorecard Rating	Actual CRISIL Rating
FY17	25	High Risk	AA-
FY18	30	High Risk	AA-
FY19	27	High Risk	AA-
FY20	27	High Risk	AA
FY21	28	High Risk	AA
FY22	31	High Risk	AA
FY23	37	High Risk	AA
FY24	49	Weak	AA+
FY25	59	Moderate	AA+
FY26	38	High Risk	AA+

The scorecard classifies Tata Power as **"High Risk" in 7 of 10 fiscal years**, while the company's actual CRISIL rating rose steadily and monotonically from AA- to AA+ with **zero downgrades** over the identical period. This is a genuine and informative finding rather than a flaw: the scorecard is deliberately built as a purely quantitative framework, and by design excludes qualitative business-risk factors — regulatory framework strength (India's assured-RoE tariff regime), Tata Group parent-company support, and management's demonstrated execution on deleveraging and portfolio diversification — that real rating agencies weight heavily and that materially support Tata Power's actual investment-grade rating. The divergence illustrates a structural property of bottom-up quantitative scorecards: they are conservative by construction relative to agency ratings that also incorporate forward-looking and qualitative judgment.

A secondary, sharper observation: the composite score falls from **59 (FY25) to 38 (FY26)** — a 21-point single-year decline — while CRISIL's actual rating held flat at AA+/Stable with no rating action. This is driven primarily by **Net Debt/EBITDA crossing a scoring-band threshold (3.98x → 5.15x)** as total debt rose from ₹62,866 Cr to ₹76,141 Cr to fund the accelerating renewable capex cycle — a move CRISIL's own published rating rationale treats as capex-funded and temporary rather than a genuine credit deterioration, consistent with their stated medium-term leverage guidance of sub-4.0x Net Debt/EBITDA.

8. Altman Z-Score — Legacy Cross-Reference

The Altman Z-Score is retained in this workbook as a deliberate methodological contrast — showing what the standard model says, alongside why it should not be trusted for this company — rather than as a genuine input to the credit assessment.

Fiscal Year	FY17	FY18	FY19	FY20	FY21	FY22	FY23	FY24	FY25	FY26
Z-Score	0.86	0.92	0.94	0.78	0.96	1.44	1.32	2.04	1.81	1.55
Classification	Distress	Distress	Distress	Distress	Distress	Distress	Distress	Grey Zone	Grey Zone	Distress

The Z-Score classifies Tata Power as **"Distress" in 8 of 10 fiscal years** despite the company's investment-grade CRISIL AA+/Stable and ICRA AA+/Stable ratings. This divergence directly motivated the construction of the Utility Credit Quality Scorecard in Section 7, and the Altman Z-Score is retained in the composite scorecard only at a deliberately down-weighted 5-point maximum, reflecting its acknowledged unsuitability rather than genuine predictive value for this sector.

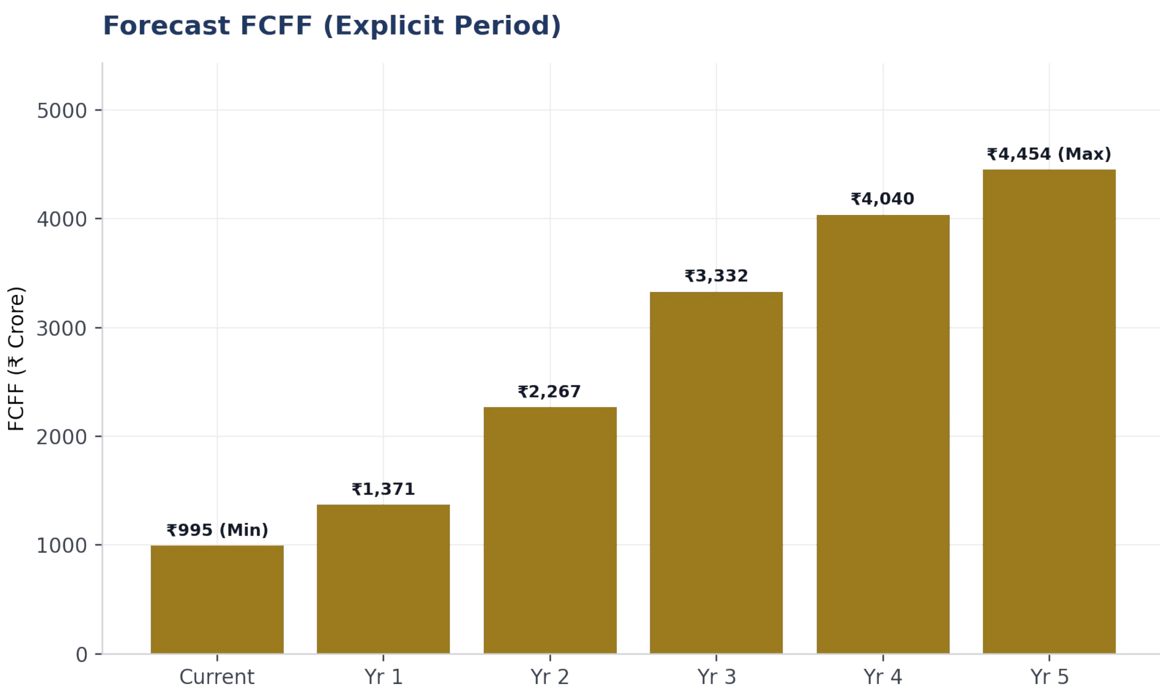
9. Discounted Cash Flow Valuation

Key Assumptions

- 3-stage Free Cash Flow to Firm (FCFF) model, explicit 6-year forecast (FY27–FY32) plus terminal value
- Growth linked to reinvestment via g = ROIC × Reinvestment Rate
- Reinvestment rate declines from 84% to 56% of NOPAT across the forecast period, reflecting Tata Power's active renewable capex buildout funded substantially by debt/equity issuance rather than retained NOPAT
- Terminal growth rate of 4.5%, constrained below the 6.8% risk-free rate per Damodaran's stable-growth discipline
- Mid-year discounting convention applied throughout the explicit forecast period

Forecast Free Cash Flow to Firm (FCFF)

FCFF grows from ₹995 Cr in the current year to **₹4,910 Cr by Year 6** of the explicit forecast, as the reinvestment rate steadily declines from 84% toward 56% of NOPAT under the Base Case (the scenario actually used in this valuation — see the three-scenario comparison immediately below).



Reinvestment Rate Scenarios — All Three Cases

The DCF model includes three distinct reinvestment-rate scenarios, selectable via a scenario toggle. The Base Case (used throughout this report) is the selected case; Best and Worst cases are presented here for completeness, since a reinvestment-rate assumption this influential warrants showing the full range considered, not just the case selected.

Scenario	Yr 1	Yr 2	Yr 3	Yr 4	Yr 5	Yr 6
Best Case	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%
Base Case (Selected)	80.0%	70.0%	60.0%	56.0%	56.0%	56.0%
Worst Case	95.0%	85.0%	80.0%	76.0%	76.0%	76.0%

Reinvestment Rate as % of NOPAT. Base Case anchored to the historical 4-year median (~157%) declining toward a more sustainable long-run rate; Best/Worst cases bound the range.

Holding all other assumptions constant (WACC, terminal growth, cash/debt/investments) and recomputing the full DCF bridge under each scenario's reinvestment-rate path gives the following equity values:

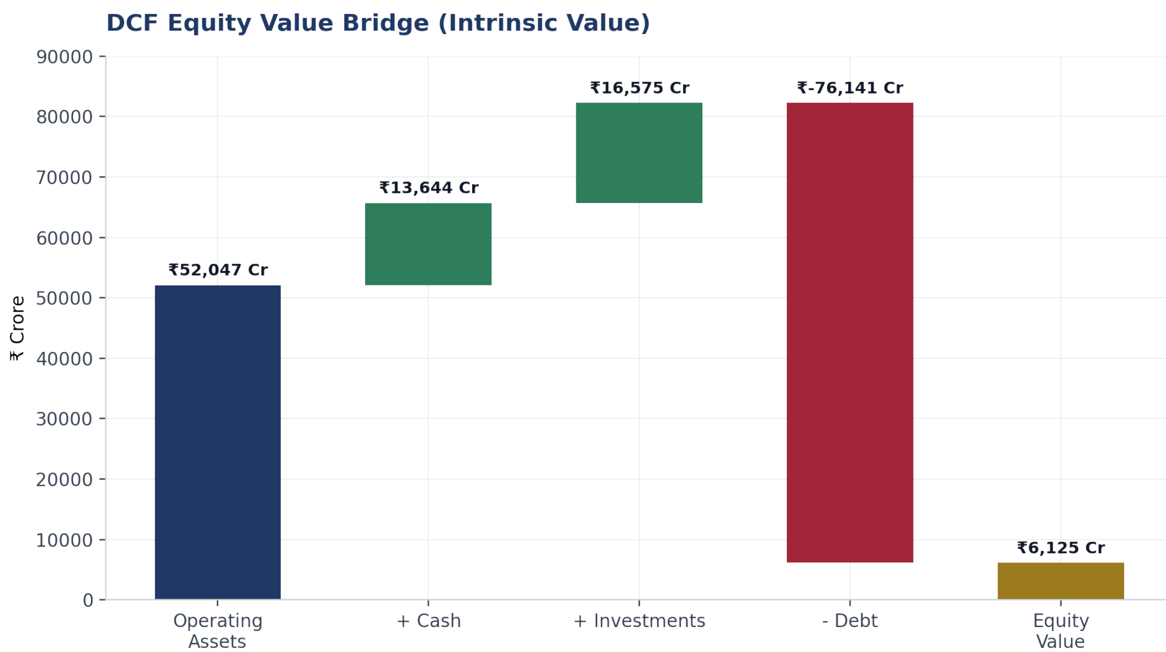
Scenario	Op. Assets (₹ Cr)	Equity Value (₹ Cr)	Value / Share	vs. Market
Best Case	74,655	28,733	₹89.92	-77.7%

Base Case (Selected)	52,047	6,125	₹19.17	-95.2%
Worst Case	27,806	(18,116)	(₹56.70)	n/m

Even under the **Best Case** (a flat 40% reinvestment rate, well below the historical median), the DCF value of **₹89.92/share remains 77.7% below market price —** reinforcing that the valuation gap identified in this report is not primarily a function of which reinvestment scenario is selected. The **Worst Case produces a negative equity value**, a mathematically valid but economically extreme output of the model under a sustained ~80%+ reinvestment rate combined with the current debt load; it is presented for completeness rather than as a realistic base expectation.

Valuation Bridge

Component	₹ Crore
PV of Forecast FCFF	13,833
PV of Terminal Value	38,214
Value of Operating Assets	52,047
Add: Cash	13,644
Add: Investments (non-operating)	16,575
Less: Debt	(76,141)
Value of Equity	6,125
Number of Shares (Cr)	319.53
Equity Value per Share (Intrinsic Value)	₹19.17
Market Price per Share	₹402.35
Discount to Market Price	-95.24%



Sensitivity Analysis — Equity Value per Share (₹)

The DCF output is highly sensitive to the WACC / terminal growth assumption pair. The table below shows equity value per share across a WACC range of 8.55%–14.55% and a terminal growth range of 2.5%–6.5%, with the base case (WACC 12.55%, terminal growth 4.5%) highlighted.

Growth ↓ / WACC →	8.55%	10.55%	12.55%	14.55%
2.5%	₹40.66	₹5.61	(₹15.49)	(₹29.58)
3.5%	₹70.24	₹21.83	(₹5.18)	(₹22.42)
4.5% (Base)	₹114.43	₹43.41	**₹19.17**	(₹13.83)
5.5%	₹187.58	₹73.54	₹24.20	(₹3.34)
6.5%	₹332.05	₹118.54	₹46.17	₹9.75

Green = higher implied value; red = negative implied value. Gold cell marks the base case.

The sensitivity grid shows that **even under the most favorable combination tested (WACC 8.55%, terminal growth 6.5%), the DCF value (₹332.05) remains just below the current market price of ₹402.35** — indicating the gap identified in this analysis is not merely a base-case artifact of one aggressive assumption, but is structurally embedded across a wide range of reasonable WACC/growth combinations. This strengthens the case that the primary explanation lies elsewhere — in the reinvestment-rate profile and the unmodeled TPREL growth-option value — rather than in the terminal-value assumptions alone.

Explaining the Gap

A DCF value **95% below market price** is a large divergence that warrants direct explanation rather than silent presentation. Two drivers were identified. First, **the reinvestment-rate assumption** — anchored to a historical 4-year median reinvestment rate of approximately 157% of NOPAT, itself a reflection of a company investing more than its entire after-tax operating profit during an active capex cycle — substantially suppresses near-term free cash flow even as it declines toward a more sustainable 56% by the end of the explicit forecast. Second, and structurally more significant: as a consolidated single-entity FCFF model, this DCF **does not separately capture the value of Tata Power Renewable Energy Ltd. (TPREL)**, which is not separately listed and may carry meaningful growth-option value that the market prices into the parent company's shares. A sum-of-the-parts (SOTP) valuation — separately valuing the regulated generation/T&D business and the renewables growth business — was identified as the methodologically appropriate extension to close this gap, but was outside this project's scope.

A Further Limitation: The Model Assumes Smooth Growth, Not a Transition "Boom"

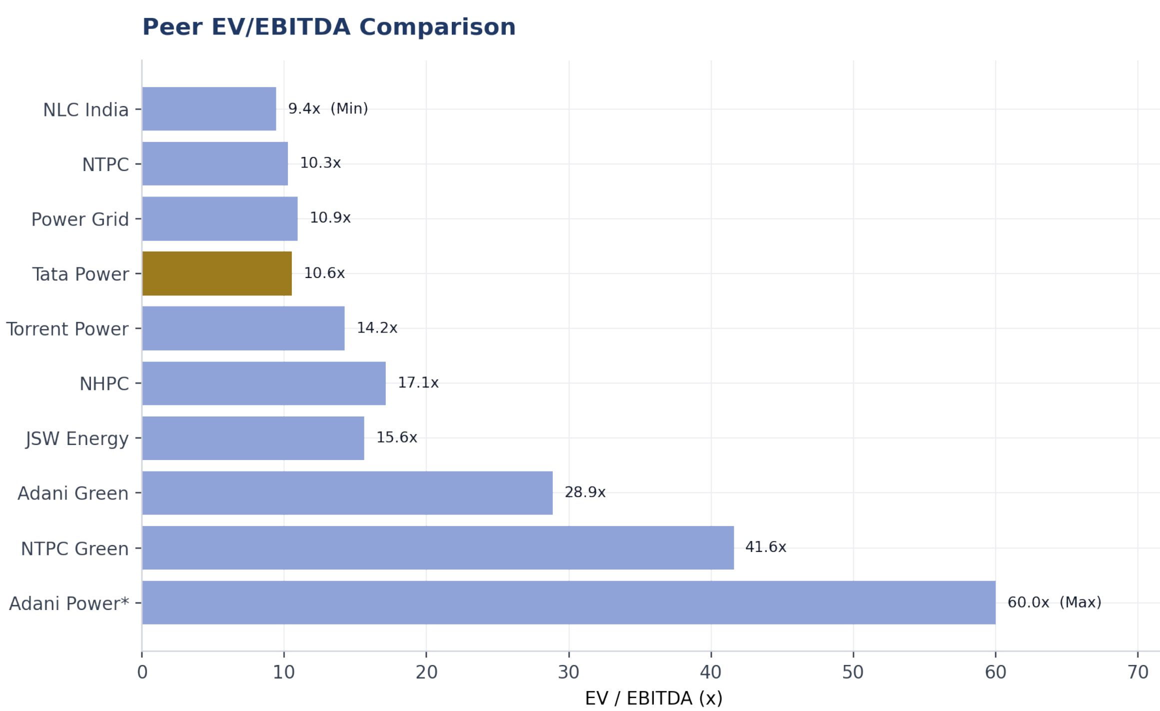
Even a lower reinvestment-rate assumption — one aggressive enough to push the DCF to a positive equity value — does not fully resolve the underlying issue, because it only addresses the near-term cash flow suppression, not the **shape** of the growth path this model assumes. This DCF links growth smoothly and continuously to reinvestment (g = ROIC × Reinvestment Rate), which implicitly assumes NOPAT grows gradually, year over year, in step with a steadily declining reinvestment rate. For a business in an active capital-intensive transition — such as Tata Power's renewable buildout — **actual earnings may not follow this smooth path at all**. A more realistic alternative is a "boom" or step-change trajectory: NOPAT stays suppressed through the construction and ramp-up phase, then rises sharply once new renewable capacity is fully commissioned and contributing full-year revenue — a pattern common to capital-intensive infrastructure buildouts, but fundamentally different from the continuous growth curve this single-path DCF assumes.

Modeling that kind of scenario properly would require its own assumptions — a commissioning schedule, a ramp-up curve, and a steady-state utilization rate for the renewable portfolio — none of which this project attempted to build. **This is a real limitation of the current DCF, separate from the reinvestment-rate and SOTP points above**: it is not just that the model is conservative on cash flow timing, but that it has no mechanism to capture a plausible non-linear earnings inflection as the transition completes. Flagged here as a direction for future extension, not resolved in this version of the model.

10. Relative Valuation

Peer Trading Multiples

Company	EV/Revenue	EV/EBITDA	P/E
Adani Power	82.39x	188.64x	340.87x
NTPC	3.23x	10.26x	12.39x
Power Grid Corpn	8.63x	10.95x	16.58x
Adani Green	27.24x	28.89x	126.51x
Adani Energy Sol	8.21x	25.66x	75.70x
Tata Power Co.	3.06x	10.55x	25.12x
JSW Energy	9.33x	15.63x	38.07x
NTPC Green Ene.	39.10x	41.60x	154.71x
NHPC Ltd	11.06x	17.12x	18.90x
Torrent Power	2.91x	14.25x	28.85x
NLC India	4.08x	9.45x	11.77x



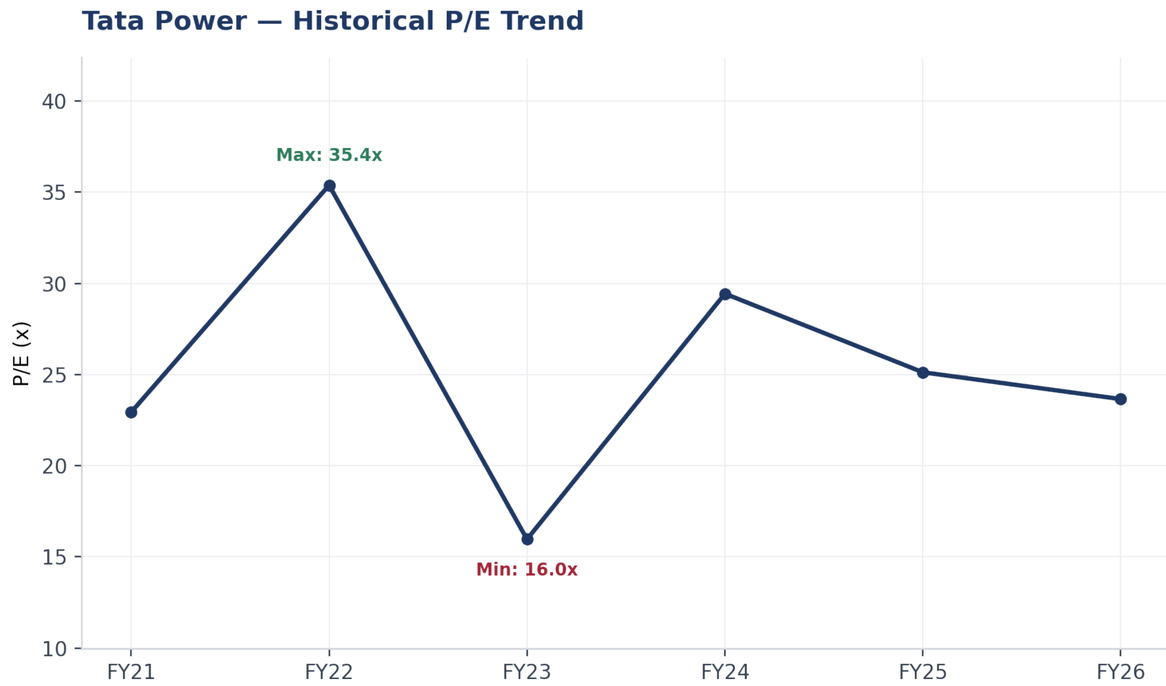
Tata Power trades at an EV/EBITDA of 10.55x, in line with mature integrated peers (NTPC at 10.26x, Power Grid at 10.95x) and well below early-stage renewable pure-plays (Adani Green at 28.89x, NTPC Green at 41.60x), whose elevated multiples reflect near-zero current earnings rather than comparable fundamentals. Adani Power's 188.64x EV/EBITDA is similarly an earnings-quality distortion rather than a meaningful comparable.

Tata Power's Own Historical Trading Multiples

Distinct from the peer comparison above, Tata Power's own trading multiples over time provide a view of how the market has re-rated the stock independent of the broader peer set.

Fiscal Year	P/E	EV/EBITDA	EV/Sales	P/B
FY21	22.93x	10.27x	2.26x	45.76x
FY22	35.38x	15.40x	2.82x	45.52x
FY23	15.96x	12.02x	1.86x	35.49x
FY24	29.43x	15.01x	2.78x	31.57x

FY25	25.12x	11.77x	2.61x	28.50x
FY26	23.65x	13.41x	2.94x	25.88x



P/E has ranged widely (15.96x to 35.38x) over just six years, reflecting both earnings volatility (the FY22 spike to 35.38x coincides with the margin-compression trough in Section 3) and shifting market sentiment on the renewable transition story. Price-to-Book has declined steadily from 45.76x to 25.88x — not because the stock has fallen, but because book equity has grown faster than the share price, a pattern consistent with an improving, if still rich, valuation on a book basis.

PEG Ratio

Using FY26 P/E (23.65x) against the trailing EPS growth rate (7.17%), the PEG ratio computes to approximately **3.30x** — well above the traditional 'fair value' benchmark of 1.0x, indicating **the market is pricing in growth expectations substantially ahead of trailing EPS growth**. This is directionally consistent with the DCF and football field findings: the market appears to be pricing in a forward growth story (plausibly the renewables transition) that trailing fundamentals alone do not yet fully evidence.

Regression-Based Relative Valuation — Methodology Demonstration

A supplementary multi-variable regression of peer P/E against Return on Equity, Dividend Payout Ratio, Growth, and Beta was constructed to test whether cross-sectional fundamentals explain observed peer multiples.

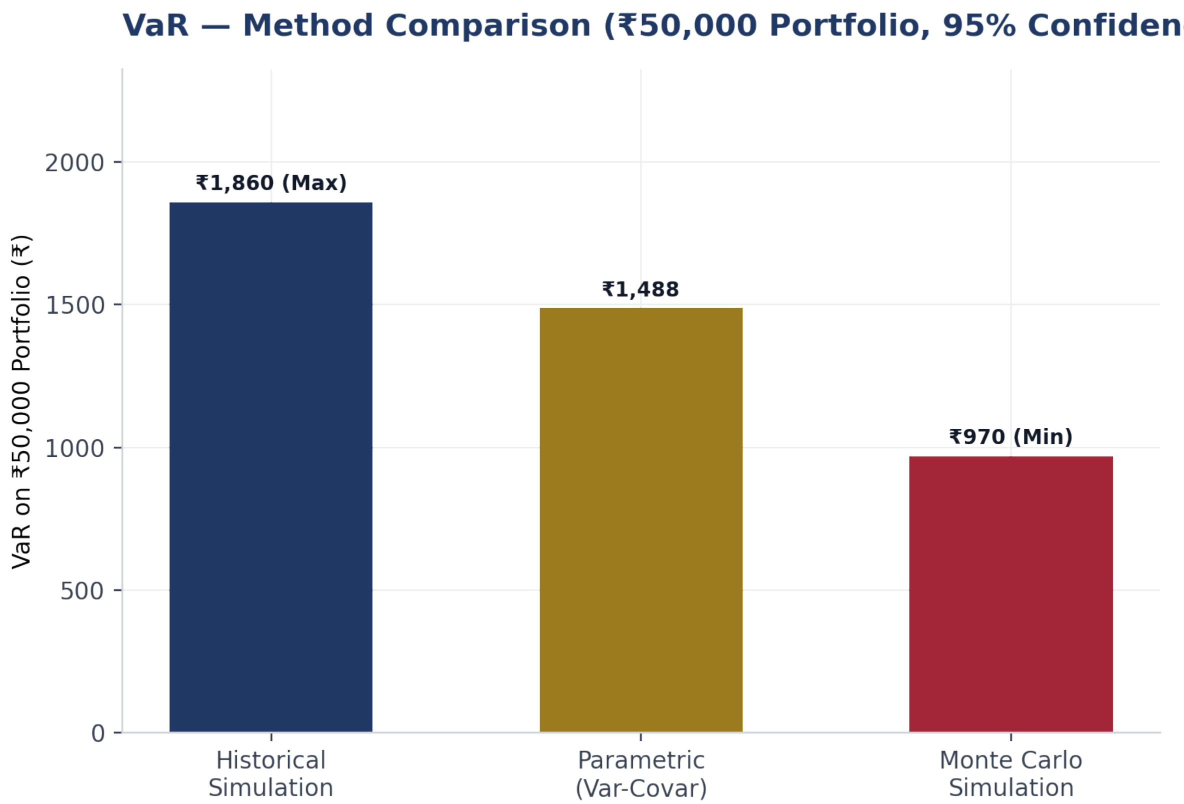
Variable	Coefficient	Standard Error	t-Stat	P-value
Intercept	45.59	27.46	1.66	0.114
ROE %	6.97	136.54	0.05	0.960
Dividend Payout %	20.05	26.62	0.75	0.461
Growth %	-359.66	395.11	-0.91	0.375
Beta (Risk)	-1.87	25.16	-0.07	0.941

$R^2 = 0.058$; Adjusted $R^2 = -0.151$ (negative, indicating the model explains no more variance than its own degrees-of-freedom penalty would predict by chance). **Every coefficient has a p-value exceeding 0.11**, with three exceeding 0.37. This regression **is not statistically significant** and its output — including the implied "regression P/E" for Tata Power — is retained solely to demonstrate multi-variable regression methodology, correct interpretation of statistical output (R^2, t-statistics, p-values), and the ability to recognize when a model fails to meet statistical thresholds. It is **explicitly excluded from the valuation conclusion**.

11. Value-at-Risk Analysis

Portfolio downside risk was assessed for a two-asset portfolio (Tata Power and Nifty 50, 40%/60% weighting) using three independent methodologies, cross-validating results at a 95% confidence level over a 5-day horizon.

Method	Result	Basis
Historical Simulation VaR	-3.72% (≈ -₹1,860 on ₹50,000)	Empirical distribution of actual observed portfolio returns
Parametric (Variance-Covariance) VaR	-₹1,488	On a ₹50,000 portfolio position; assumes normally distributed returns
Monte Carlo Simulation VaR	-₹970	On a ₹50,000 portfolio position; 10,000+ simulated price paths using estimated variance and covariance

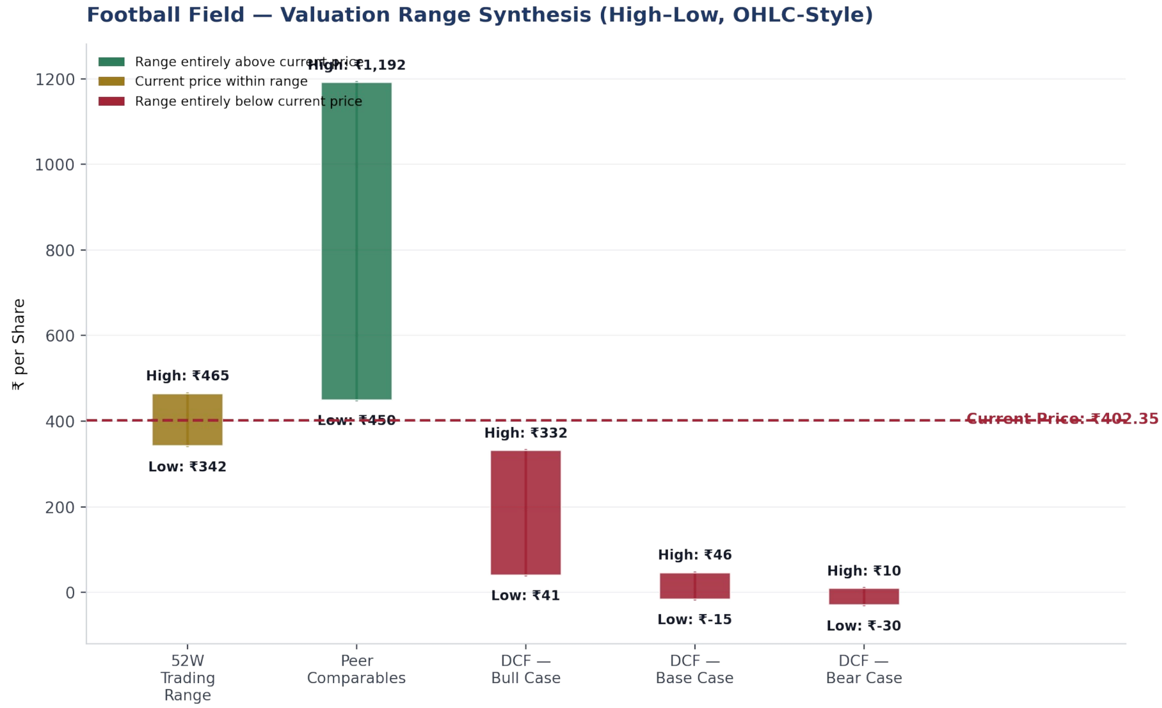


The parametric and Monte Carlo estimates diverge meaningfully in magnitude (₹1,488 vs. ₹970), which is expected given their different distributional assumptions — the parametric method assumes normality and can overstate tail risk for assets with actual return distributions that are less extreme than a normal distribution would imply, while Monte Carlo simulation draws directly from estimated portfolio variance and covariance without imposing a parametric shape. Presenting all three side by side, rather than a single figure, is itself the more defensible practice: it makes the method-dependency of VaR estimates visible rather than implying false precision.

12. Football Field Synthesis

The football field consolidates all valuation approaches into a single comparative view:

Method	Low (₹/share)	High (₹/share)
52-Week Trading Range	342.35	464.90
DCF — Bear Case	(29.58)	9.75
DCF — Base Case	(15.49)	46.17
DCF — Bull Case	40.66	332.05
Peer Comparables (EV/EBITDA-implied)	449.51	1,191.50



The current market price of ₹402.35 **sits above every DCF scenario** (including the Bull case) but within the peer comparables range and just below the 52-week trading range high. This pattern is directionally consistent with the findings in Sections 9 and 10: the DCF's consolidated single-entity structure and aggressive reinvestment assumptions produce a materially more conservative valuation than either the market or peer-multiple-based approaches, reinforcing the case for a sum-of-the-parts extension as the primary avenue for narrowing this range in future work.

13. Limitations & Disclosures

- **DCF vs. market price gap:** The ~95% discount to market price is attributed to reinvestment-rate assumptions and unmodeled subsidiary/growth-option value (TPREL). A sum-of-the-parts valuation was outside this project's scope.
- **No modeled “boom” scenario for NOPAT:** The DCF assumes smooth, continuous growth (g = ROIC × Reinvestment Rate). It does not model a step-change/non-linear NOPAT trajectory that could plausibly follow once Tata Power's renewable transition completes and new capacity reaches full-year utilization — a real limitation, separate from the reinvestment-rate and SOTP points above.
- **Regression-based relative valuation:** $R^2 = 0.058$, Adjusted R^2 negative, all coefficients statistically insignificant ($p > 0.11$). Retained solely to demonstrate methodology; explicitly excluded from the valuation conclusion.
- **Altman Z-Score:** The 'Distress' classification in 8 of 10 years is a model-fit artifact for a capital-intensive regulated utility, not a genuine credit signal — corroborated by the company's AA+ ratings and the custom scorecard built to replace it.
- **Utility Credit Quality Scorecard:** A purely quantitative framework; it excludes qualitative business-risk factors (regulatory framework strength, Tata Group parent support, management execution) that materially support the company's actual investment-grade rating. This explains why the scorecard runs structurally more conservative than CRISIL's own assessment.
- **Equity Risk Premium:** Historical-returns-based ERP for India computes to a negative figure over the sample window and is not used; a rating-based ERP is used instead, consistent with Damodaran's guidance for emerging markets.
- **FFO approximation:** Funds From Operations throughout this analysis is approximated as Net Profit + Depreciation, a standard simplification used in the absence of a full cash-flow-statement breakout of non-cash and working-capital items.

14. Conclusion

This project was built as a demonstration of end-to-end equity research methodology — financial statement analysis, risk modeling, cost of capital construction, discounted cash flow valuation, relative valuation, and a custom sector-appropriate credit scoring framework — drawing selectively on frameworks from Aswath Damodaran's valuation teaching, adapted throughout to the data actually available rather than a full replication of his methodology. Where the analysis produced results that diverge from market pricing or from textbook model assumptions, those divergences were investigated and disclosed rather than smoothed over: the Altman Z-Score's mismatch for a regulated utility motivated the construction of a custom credit scorecard; the DCF's large discount to market price is explained by identified, named drivers rather than left unexplained; and a statistically insignificant regression is presented transparently as a methodology demonstration rather than a valuation input. This report and its underlying workbook are intended for educational and portfolio purposes only and do not constitute investment, financial, or legal advice.

Sources: Company financial statements (via Screener.in), Tata Power investor presentations (FY25–FY26), CRISIL rating rationales (2019–2025). Valuation approach informed by Aswath Damodaran (NYU Stern), adapted throughout to the data available.